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Exercises are from Stephen Abbott's *Understanding Analysis*, Second Edition.

Problem 1: Subsets of disjoint open sets are separated

(Exercise 3.4.5)

Proof idea

This proof follows just from messing around a little with complements, and remembering that the closure of a set is the smallest closed set that contains it. The sets A and B are a bit of a distraction; what's really going on is that U and V are separated.

In my informal mental picture, there's a bit of "space" between U and V . Also, I imagined $U = (-\infty, 0)$ and $V = (0, \infty)$, like in the proof that all connected sets in $\mathbb{R}$ are intervals.

Proof

Theorem 1: Let A and B be nonempty subsets of $\mathbb{R}$. Let U and V be disjoint open sets with $A \subseteq U$ and $B \subseteq V$. Then A and B are separated.

Proof: Since U is open and disjoint from V , U^C is closed and contains V , so it contains B as well. Since $\overline{B}$ is the smallest closed set that contains B , we have $\overline{B} \subseteq U^C$. Consequently, $U \cap \overline{B} = \emptyset$. But $A \subseteq U$, so $A \cap \overline{B} = \emptyset$ as well.

By an exactly symmetric argument, $\overline{A} \cap B = \emptyset$. Since A and B are given nonempty, these intersections show that they are separated. ■

Problem 2: Connected $\Leftrightarrow$ in every partition into two sets, one contains a limit point of the other

(Exercise 3.4.6)

Proof idea

Just unwind definitions.

Proof

Theorem 2: A set $E \subseteq \mathbb{R}$ is connected if and only if, for all nonempty disjoint sets A and B satisfying $E = A \cup B$, there always exists a convergent sequence $(x_n) \rightarrow x$ with (x_n) contained in one of A and B , and x an element of the other.

Proof:

($\Rightarrow$)

Suppose E is connected. Then it is not disconnected. So for all nonempty disjoint sets A and B satisfying $E = A \cup B$, they are not separated. So one of $\overline{A} \cap B$ and $A \cap \overline{B}$ is nonempty.

Without loss of generality, suppose that $\overline{A} \cap B$ is nonempty. Let A' denote the set of limit points of A . Since $A \cap B = \emptyset$ and $\overline{A} = A \cup A'$, it must be that $A' \cap B$ is nonempty. That is, a limit point x of A falls in B . Since x is a limit point of A , there is a sequence (x_n) whose terms lie in A and $(x_n) \rightarrow x$, with $x \in B$.

($\Leftarrow$)

We will prove the contrapositive. Suppose E is disconnected. Then it may be written as $E = A \cup B$, where A and B are nonempty, and $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$. Then A and B must be disjoint, otherwise the closure of one would have nonempty intersection with the other.

Suppose for the sake of contradiction that a sequence (x_n) whose terms are all in A converged to a point x in B . Then, since $(x_n) \rightarrow x$ and $x_n \in A$ but $x \notin A$, the point x is a limit point of A , so $x \in \overline{A}$, contradicting that A and B are separated. So no such sequence exists.

An exactly symmetric argument shows that no sequence in B can converge to a point in A .

We have proved that E disconnected $\Rightarrow$ there exist nonempty disjoint A and B where no sequence in one converges to a point in the other. This is the required contrapositive. ■

Problem 3: $\mathbb{Q}$ is totally disconnected, and so are the irrationals

(Exercise 3.4.7)

Proof idea

Given two rational numbers, find a multiple of $\sqrt{2}$ between them. Then use the construction in the proof that all connected subsets of $\mathbb{R}$ are intervals: two open intervals with a shared endpoint.

To find the point between them, note that $1 < \sqrt{2} < 2$ and find an affine mapping that sends 1 to a and 2 to b , then take whatever it sends $\sqrt{2}$ to.

For the irrationals, the density of $\mathbb{Q}$ in $\mathbb{R}$ hands us a rational directly.

Proof

(a)

Theorem 3: $\mathbb{Q}$ is totally disconnected.

Proof: Let $x, y \in \mathbb{Q}$ be any two distinct rational numbers. Without loss of generality let $x < y$ and let

$$c = y(\sqrt{2} - 1) - x(\sqrt{2} - 2).$$

Then c is irrational, for otherwise

$$\frac{c - 2x + y}{y - x} \equiv \sqrt{2}$$

would be an algebraic combination of rational numbers and therefore rational (here we use that the field $\mathbb{Q}$ is closed under the usual operations of addition, subtraction, multiplication and division by a nonzero rational).

We claim that $c > x$ and $c < y$, for

$$c - x = (y - x)(\sqrt{2} - 1) > 0$$

and

$$c - y = (y - x)(\sqrt{2} - 2) < 0.$$

Let $A = (-\infty, c) \cap \mathbb{Q}$ and $B = (c, \infty) \cap \mathbb{Q}$. Then

$$A \cup B = ((-\infty, c) \cup (c, \infty)) \cap \mathbb{Q} = (\mathbb{R} \setminus \{c\}) \cap \mathbb{Q} = \mathbb{Q}$$

and $x \in A$ and $y \in B$.

But by the Order Limit Theorem, all limit points of A are contained in $(-\infty, c]$, and all limit points of B are contained in $[c, \infty)$. So $\overline{A} \subseteq (-\infty, c]$ while $B \subseteq (c, \infty)$ so $\overline{A} \cap B = \emptyset$ and, symmetrically, $A \cap \overline{B} = \emptyset$. Therefore A and B are separated. Since x and y were arbitrary, we have proved $\mathbb{Q}$ to be totally disconnected. ■

Remark 0.1: We don't really need the explicit construction. We could have said, By density of $\mathbb{Q}$ in $\mathbb{R}$, there's a nonzero rational d between $\frac{x}{\sqrt{2}}$ and $\frac{y}{\sqrt{2}}$. Then set $c = \sqrt{2}d$.

(b)

Theorem 4: The set of irrational numbers is totally disconnected.

Proof: Let $x, y \in \mathbb{R} \setminus \mathbb{Q}$ be any two distinct irrational numbers. Without loss of generality let $x < y$. Since they are real and by the density of $\mathbb{Q}$ in $\mathbb{R}$, there exists a $q \in \mathbb{Q}$ such that $x < q < y$.

Let $A = (-\infty, q) \cap (\mathbb{R} \setminus \mathbb{Q})$ and $B = (q, \infty) \cap (\mathbb{R} \setminus \mathbb{Q})$. Then

$$A \cup B = ((-\infty, q) \cup (q, \infty)) \cap (\mathbb{R} \setminus \mathbb{Q}) = (\mathbb{R} \setminus \{q\}) \cap (\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R} \setminus \mathbb{Q}$$

and $x \in A$ and $y \in B$.

By the same finishing argument as in part (a), A and B are separated and so the irrationals are totally disconnected. ■

Problem 4: Connected components

(a)

Proof idea

If the union is disconnected then we get separated sets A and B which we can intersect with each member of the collection. This forces each member to live in exactly one of A or B . Both must contain at least *one* member each in order to be nonempty. But that leaves nowhere for x to go.

Proof

Theorem 5: Let $\mathcal{S}$ be a collection of connected sets and suppose that there is a point x which is in every set in $\mathcal{S}$. Then $\bigcup \mathcal{S}$ is connected.

Proof: Let S_α denote a member of $\mathcal{S}$ and

$$U = \bigcup S_\alpha$$

and x some point such that $x \in S_\alpha$ for every $S_\alpha \in \mathcal{S}$.

Suppose for the sake of contradiction that U is disconnected. Then $U = A \cup B$ where A and B are nonempty and separated. So x must be in exactly one of A or B . Without loss of generality, let $x \in A$.

For every $S_\alpha \in \mathcal{S}$ write

$$S_\alpha = S_\alpha \cap U = (S_\alpha \cap A) \cup (S_\alpha \cap B) = A_\alpha \cup B_\alpha$$

where

$$A_\alpha = S_\alpha \cap A$$

and

$$B_\alpha = S_\alpha \cap B.$$

Since $\overline{A} \cap B = \emptyset$, and $\overline{A} = A \cup A'$ where A' is the set of limit points of A , and $A'_\alpha \subseteq A'$, we see that

$$\overline{A_\alpha} \subseteq \overline{A},$$

so

$$\overline{A_\alpha} \cap B = \emptyset,$$

so

$$\overline{A_\alpha} \cap B_\alpha = \emptyset.$$

By a symmetric argument,

$$A_\alpha \cap \overline{B_\alpha} = \emptyset.$$

But each S_α is connected, so one of A_α and B_α must be empty, and $S_\alpha = A_\alpha$ or $S_\alpha = B_\alpha$, which means S_α is entirely contained within one of A or B .

If any $S_\alpha \subseteq B$, then $x \notin S_\alpha$ since $x \in A$, a contradiction. But if every $S_\alpha \subseteq A$ then B is empty, a contradiction. ■

(b)

Theorem 6: Let $A \subseteq \mathbb{R}$ and for $x, y \in A$ define the relation $\sim$ by declaring that $x \sim y$ if there is a connected set $C \subseteq A$ containing both x and y . Then this is an equivalence relation.

Proof: Reflexive: $x \sim x$ since the singleton set $\{x\}$ is connected: it cannot be written as the union of two separated sets because it cannot be written as the union of two nonempty sets at all.

Symmetric: if $x \sim y$ then there is a connected set $C \subseteq A$ containing both x and y , so $y \sim x$.

Transitive: if $x \sim y$ and $y \sim z$ then there are connected sets $C \subseteq A$ and $D \subseteq A$, where $x, y \in C$ and $y, z \in D$. Notice that y is in both C and D , so we may apply the theorem in part (a) to the collection $\{C, D\}$ to obtain that $C \cup D$ is connected. Now, $x, z \in C \cup D$ and $C \cup D \subseteq A$ so $x \sim z$. ■

The equivalence classes with respect to this relation are called the **connected components** of A .

(c)

Theorem 7: If C is the connected component of some $a \in A$, then C is the largest connected subset of A containing a (so in particular C is connected), and there exists a closed set $C' \subseteq \mathbb{R}$ such that $C = C' \cap A$ (that is, C is “closed in A ”).

Proof: Let $B \subseteq A$ be connected subset of A such that $a \in B$. Let $y \in B$ be any point of B . Then, since the connected set B contains both a and y , we have $a \sim y$. But because C is defined to be the equivalence class of a with respect to $\sim$, we have $y \in C$. Therefore $B \subseteq C$.

Now suppose for the sake of contradiction that C is disconnected. Then $C = C_1 \cup C_2$ where C_1 and C_2 are nonempty separated sets. Let $x \in C_1$ and $y \in C_2$. Then, since C is defined as an equivalence class of $\sim$, we have $x \sim y$, so some connected $D \subseteq A$ contains both x and y . We proved above that $D \subseteq C$. Define

$$\begin{aligned} D_1 &\equiv D \cap C_1 \\ &\text{and} \\ D_2 &\equiv D \cap C_2 \end{aligned}$$

whereby $x \in D_1$ and $y \in D_2$ so they are nonempty.

Since $D \subseteq C$ we may write

$$D = D \cap C = D \cap (C_1 \cup C_2) = D_1 \cup D_2.$$

But since $\overline{C_1} \cap C_2 = \emptyset$, we have $\overline{D_1} \cap D_2 = \emptyset$, and similarly $D_1 \cap \overline{D_2} = \emptyset$, making D disconnected, a contradiction. Therefore C is connected.

Now we find a closed set $C' \subseteq \mathbb{R}$ such that $C = C' \cap A$. Take $C' = \overline{C}$. It is clear that $C \subseteq \overline{C} \cap A$. We claim that $\overline{C} \cap A \subseteq C$:

Let $x \in \overline{C} \cap A$. If $x \in C$ then we are done. Otherwise $x \in L \cap A$ where L is the set of limit points of C . We claim that $C \cup \{x\}$ is connected. Suppose for the sake of contradiction that it is not. Then $C \cup \{x\} = E \cup F$ where E and F are nonempty and separated. Since C itself is connected, all of it must lie entirely within one of E or F (otherwise, $C \cap E$ and $C \cap F$ would be nonempty and separated, showing C to be disconnected.) Without loss of generality let $C \subseteq E$. Then $F = \{x\}$ is the only way for F to be nonempty. But now, $\overline{C} \cap F = \{x\} \neq \emptyset$ since $x \in \overline{C}$. Therefore $\overline{E} \cap F \neq \emptyset$. So $C \cup \{x\}$ is connected.

But $C \cup \{x\} \subseteq A$, so as a connected subset of A containing points of C , it must be a subset of C , as proved above. So $x \in C$, and therefore $\overline{C} \cap A \subseteq C$. We have shown both set inclusions and so conclude that

$$C = \overline{C} \cap A.$$

■

In response to Nic's email, here's a simpler proof that C is connected:

Theorem 8: C is connected.

Proof: C is the equivalence class of a , so $a \in C$. And for every $x \in C$ we have $x \sim a$, so there exists some connected set $D_x \subseteq A$ that contains both x and a . By the argument before (that every connected subset of A that contains a is a subset of C), $D_x \subseteq C$. Therefore

$$C = \bigcup_{x \in C} D_x$$

where $a \in D_x$ for each D_x containing an $x \in C$. By the previous result, C is connected. ■

And a simpler proof that $\overline{C} \cap A = C$:

Lemma 1: Let X be any connected subset of A . Then $\overline{X} \cap A$ is connected.

Proof: Write $Y = \overline{X} \cap A = X \cup (X' \cap A)$ where X' is the set of limit points of X .

We will show that in every partition of $Y = E \cup F$ into two nonempty disjoint sets, one contains a limit point of the other.

Suppose that some points of X are in E , while others are in F . Then since X is connected we are done: we know that one of $X \cap E$ and $X \cap F$ must contain a limit point of the other.

Otherwise suppose that all of X falls in one partition, say E . Then at least one point of X' has to go in F in order for F to be nonempty. But then F contains a limit point of E . So Y is connected. ■

Theorem 9: $\overline{C} \cap A = C$.

Proof: By the lemma above, since C is connected, $\overline{C} \cap A$ is connected. Since it is a connected subset of A that contains a , it must be a subset of C . But it contains all of C . So it just equals C . ■

(d)

The connected components of $(1, 2) \cup (2, 3)$ are $(1, 2)$ and $(2, 3)$: as intervals they are connected, and they're separated since $\overline{(1, 2)} \cap (2, 3) = (1, 2) \cap \overline{(1, 2)} = \emptyset$.

$\mathbb{Q}$ is totally disconnected, so its connected components are the singleton sets $\{q\}$ for every $q \in \mathbb{Q}$.

(e)

Theorem 10: If A has finitely many connected components, then for each connected component C , there exists an open set $C'' \subseteq \mathbb{R}$ such that $C = C'' \cap A$.

Proof: Let $\mathcal{S}$ be the collection of all the *other* connected components of A besides C , and write M_i for any member of $\mathcal{S}$.

Since every point of A is in some equivalence class (if only a singleton containing itself), the equivalence classes partition the set, so for the particular component C we can write

$$\begin{aligned} C &= A \setminus \left(\bigcup \mathcal{S} \right) \\ &= A \cap \left(\bigcup \mathcal{S} \right)^C \\ &= A \cap \left(\bigcap M_i^C \right). \end{aligned}$$

By the final conclusion of (c), each $M_i = M'_i \cap A$ where $M'_i \subseteq \mathbb{R}$ is closed. So

$$\begin{aligned} C &= A \cap \left(\bigcap (M'_i \cap A)^C \right) \\ &= A \cap \left(\bigcap (M_i'^C \cup A^C) \right) \\ &= A \cap \left(A^C \cup \left(\bigcap M_i'^C \right) \right) \\ &= (A \cap A^C) \cup \left(A \cap \left(\bigcap M_i'^C \right) \right) \\ &= A \cap \left(\bigcap M_i'^C \right) \\ &= A \cap \left(\bigcap M_i'' \right) \\ &= A \cap C'' \end{aligned}$$

where each $M_i'' \equiv M_i'^C$ is open since M'_i is closed, and

$$C'' \equiv \bigcap M_i''$$

where the intersection is finite, so $C'' \subseteq \mathbb{R}$ is open. ■

The finiteness assumption is necessary: take $A = \mathbb{Q}$ which has infinitely many connected components. Every connected component of it consists of the singleton set $\{q\}$ where $q \in \mathbb{Q}$. But there is no open set C'' in $\mathbb{R}$ such that

$$\{q\} = \mathbb{Q} \cap C''$$

because every open set in $\mathbb{R}$ contains an open ball, and by the density of $\mathbb{Q}$ in $\mathbb{R}$, every ball contains an infinite number of points of $\mathbb{Q}$.